

SOFTWARE REQUIREMENTS SPECIFICATION DOCUMENT

AI-Based Deepfake Interaction System For Entertainment

CENG 384 – Introduction to Signals Processing for Computer Engineers

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1. INTRODUCTION

1.1. Purpose of the System

The purpose of this system is to develop an AI-based deepfake interaction platform for entertainment. The system aims to create realistic and interactive virtual experiences by combining voice cloning, face swapping, and AI-driven conversation technologies. In this way, users can engage with AI-generated characters in a more dynamic and immersive manner.

This project addresses the growing interest in AI-generated media and human-computer interaction by allowing users to experience parody-based simulations of public figures in a controlled environment. It also includes evaluation features to measure the quality and performance of the generated audio, video, and real-time interaction outputs. These features help assess how effectively the system performs while supporting a more structured and responsible use of AI-generated content.

1.2. Scope of the System

The system provides an integrated platform that allows users to interact with AI-generated characters through voice, video, and text-based communication. It includes functionalities such as text-to-speech voice synthesis, optional real-time voice conversion, AI-based face swapping in video, and personality-driven conversation simulation.

Additionally, the system simulates a virtual meeting environment where users can engage in predefined scenarios such as interviews, debates, and entertainment sessions. The platform includes safety mechanisms such as visual and audio watermarking to ensure ethical usage.

The system is intended for educational, research, and entertainment purposes only and does not support unauthorized identity replication or real-world impersonation. Advanced features such as real-time processing and multi-character interaction may be included as optional components depending on system feasibility.

2. SYSTEM OVERVIEW

2.1. Overall Functionality of the System

The AI-Based Deepfake Interaction System for Entertainment is an interactive multimedia platform that enables users to communicate with AI-generated virtual characters in real time. The system combines voice cloning, face swapping, and AI-driven conversation generation to create realistic and engaging interactions. Users can provide input in text or voice form, and the system processes this input to generate synchronized audio and video outputs for an immersive deepfake interaction experience. The system also supports real-time and near real-time communication in simulated meeting environments.

2.2. Major Components or Modules of the System

The system consists of several major components that work together to provide the intended functionality. The User Interface (UI) allows users to provide input and interact with the system. The AI Conversation Module generates context-aware responses based on predefined personality models. The Voice Cloning Module converts generated responses into synthetic speech with adjustable features such as pitch, tone, and speaking rate. The Face Swap Module processes video data by detecting faces, aligning facial landmarks, and replacing the target face while preserving expressions and visual consistency. In addition, the Backend Server manages data processing, synchronization, and communication between system modules.

2.3. Main Objectives and Capabilities of the System

The main objective of the system is to provide an entertaining and interactive platform where users can experience AI-generated parody-based simulations of public figures. The system is capable of generating personality-based conversations, synthesizing speech in selected voice styles, and creating face-swapped video outputs that maintain realism in terms of facial expressions, head pose, and lighting. It also supports virtual meeting simulations with predefined scenarios such as interviews, debates, and entertainment-based conversations. Furthermore, the system includes ethical safeguards such as visible labels and inaudible watermarking to promote transparency and responsible use.

2.4. Operating System

The system is designed to operate on standard computing platforms, including Windows and macOS, and can also be supported through web-based environments depending on implementation. This allows the platform to be accessible to a wide range of users while supporting interactive multimedia processing.

3. FUNCTIONAL REQUIREMENTS

1- Voice Cloning Module

FR-1.1: The system shall provide a selection of pre-trained celebrity voice models.

FR-1.2: The system shall use publicly available audio datasets to train the voice models.

FR-1.3: The system shall convert user-provided text into synthesized speech using the selected voice model.

FR-1.4: The system shall allow the user to adjust the speaking rate of the synthesized speech.

FR-1.5: The system shall allow the user to adjust the pitch of the synthesized speech.

FR-1.6: The system may allow the user to adjust the emotional tone (e.g., happy, serious) of the synthesized speech.

FR-1.7: The system may accept live microphone input from the user.

FR-1.8: The system may convert a user's live voice into the selected celebrity voice in real-time.

FR-1.9: The system shall export the generated audio as downloadable WAV or MP3 files.

FR-1.10: The system shall embed an inaudible watermark into the generated audio output.

2- AI Video Face Swap Module

FR-2.1: The system shall include pre-trained face models of predefined public figures.

FR-2.2: The system shall use publicly available images to train the predefined face models.

FR-2.3: The system shall detect the user's face from an uploaded video or a live webcam feed.

FR-2.4: The system shall align facial landmarks prior to executing the face swap.

FR-2.5: The system shall replace the user's face with the selected celebrity face model.

FR-2.6: The system shall preserve the user's original facial expressions in the output video.

FR-2.7: The system shall preserve the user's original head pose in the output video.

FR-2.8: The system may adjust the swapped face to maintain lighting consistency with the original video.

FR-2.9: The system shall provide a simulated video meeting interface.

FR-2.10: The system may render the face-swapped output in real-time or near real-time.

FR-2.11: The system shall embed a permanent visual watermark (e.g., "AI Generated" or "Parody Mode") on the video interface.

3- AI Conversation Module

FR-3.1: The system shall generate text responses that simulate the conversational style of the selected public figure.

FR-3.2: The system shall provide a text-based chat interface for user interaction.

FR-3.3: The system may support voice-based interaction for live dialogue.

FR-3.4: The system shall log all conversation data generated during the interaction.

FR-3.5: The system shall store the conversation logs securely.

4- Meeting Simulation Feature

FR-4.1: The system shall display the AI-generated character as a visual participant within the virtual meeting room.

FR-4.2: The system shall allow the user to select a predefined "Comedy interview" scenario.

FR-4.3: The system shall allow the user to select a predefined "Job interview parody" scenario.

FR-4.4: The system shall allow the user to select a predefined "Friendly debate" scenario.

FR-4.5: The system shall allow the user to select a predefined "Motivational speech" scenario.

FR-4.6: The system may allow multiple AI-generated participants to join a single meeting session.

5- Quantitative Evaluation (Academic Requirement)

FR-5.1: The system shall calculate the Mel-Cepstral Distortion (MCD) metric for the generated voice.

FR-5.2: The system shall calculate the Signal-to-Noise Ratio (SNR) metric for the generated voice.

FR-5.3: The system shall calculate the Structural Similarity Index (SSIM) metric for the generated video.

FR-5.4: The system shall calculate the Peak Signal-to-Noise Ratio (PSNR) metric for the generated video.

FR-5.5: The system may measure the end-to-end latency during real-time simulations.

6- User Interface Requirements

FR-6.1: The system shall provide a user interface panel to select a voice model.

FR-6.2: The system shall provide a user interface panel to select a face model.

FR-6.3: The system shall provide a user interface panel to select a conversation style.

FR-6.4: The system may display a real-time visual preview before generating the final video output.

FR-6.5: The system shall provide a download function for the final interaction results.

FR-6.6: The system shall block direct integration with public auto-sharing platforms.

4. NON-FUNCTIONAL REQUIREMENTS

1- Performance

NFR-1.1: The system shall provide near-real-time interaction to support a smooth user experience.

NFR-1.2: The system shall aim to maintain responsive performance during video processing, depending on hardware capabilities.

NFR-1.3: Text-to-Speech (TTS) generation shall be completed within a reasonable time after receiving input.

NFR-1.4: The system shall maintain acceptable end-to-end latency during interaction and simulation processes.

2- Usability

NFR-2.1: The user interface shall be simple, intuitive, and accessible to non-technical users.

NFR-2.2: The system shall provide a responsive live preview with minimal delay before generating final outputs.

3- Security

NFR-3.1: The system shall ensure that generated content is clearly identifiable as artificial through visible indicators.

NFR-3.2: The system shall restrict automatic sharing of generated content to external public platforms.

NFR-3.3: User data, including voice recordings and interaction logs, shall be stored securely and accessed only for authorized purposes.

4- Reliability

NFR-4.1: The system shall maintain stable operation and consistent performance during active user sessions.

NFR-4.2: The system shall provide clear and informative error messages in case of hardware or system failures.

5- Compatibility

NFR-5.1: The system shall be compatible with modern web browsers, including Google Chrome, Microsoft Edge, and Mozilla Firefox.

NFR-5.2: The system shall support devices equipped with standard webcams and microphones.

6- Maintainability

NFR-6.1: The system shall follow a modular architecture separating core components such as voice cloning, face swapping, and conversation modules.

NFR-6.2: The source code shall include sufficient documentation and comments to support future maintenance and development.

5. SYSTEM CONSTRAINTS

5.1. Budget Constraints

The project is planned to be developed primarily by using open-source tools, libraries, and frameworks in order to reduce development costs as much as possible. For the same reason, the use of commercial or paid software licenses is intended to be avoided whenever suitable free alternatives are available. In addition, the development environments, tools, and libraries to be used in the project are expected to be selected from widely accessible and cost-free options so that the system remains economically feasible throughout the development process.

5.2. Hardware Constraints

The system shall operate on standard personal computers without requiring specialized or high-cost hardware.

The minimum hardware requirements shall be defined as follows:

- Processor: Dual-core CPU or higher
- Memory (RAM): At least 8 GB
- Storage: SSD-based storage recommended
- Additional: Standard input devices and stable internet connection

These requirements are considered sufficient for ensuring stable and efficient system performance under normal usage conditions.

5.3. Computing Capacity Constraints

Since the system includes multimedia processing and AI-based interaction features, computing capacity is an important limitation during both development and execution. The performance of the system depends on the available processing power, memory, and overall hardware capability of the device on which it runs. For this reason, the system is expected to avoid unnecessarily expensive computational operations and to use optimized processing techniques whenever possible. These limitations are important for maintaining acceptable response times, stable execution, and a consistent user experience.

5.4. Technology Choices

The technologies chosen for the project need to be widely supported, maintainable, and suitable for long-term use. Python is considered an appropriate choice for the core processing components of the system, while HTML, CSS, and JavaScript are used for user interface development. Supporting libraries such as NumPy for numerical operations, OpenCV for image processing, and Matplotlib for data visualization are also suitable within the scope of the project. Overall, the technology stack should remain stable, scalable, and manageable in order to support both development and future maintenance.